BIO 005 · YUBA COLLEGE · FALL 2026

Human Physiology.

Competency Packet · Fall 2026

15 weeks · September 8 to December 16, 2026

Section BIOL-5-D9286 · Sutter Internet (NET)

Fully online, lecture and lab, no set meeting time

For students A checklist of everything this course asks you to be able to do, and what it assumes you arrive with. Work down it. When you can do the thing without looking, that one is finished

For evaluating institutions The full assessed content of BIO 005, by unit and topic, each competency tagged for written lecture examination, laboratory work, or both, preceded by the anatomy, chemistry and quantitative skills assumed on entry

Also in Canvas The course syllabus and the course schedule

Dr. Sharilyn Rennie

What this course assumes you already have

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Twenty seven entry expectations, in three areas. They are not part of the 268 competencies below and no exam question asks them on their own. They are listed because physiology is built on them, and because a student who knows which of them is shaky can fix it before the unit that needs it arrives.

How to read this list. Each entry names the units that use it. A concept listed for more than one unit is genuinely needed more than once, at a different depth each time: pH in unit 1 is what the scale means, and pH in unit 5 is a patient with a blood gas. Every entry has a short self check on the matching unit page in the course site.

ANATOMY ASSUMED ON ENTRY

Physiology asks how things work, so it assumes you know what the things are. These are not taught here. Each one names the unit that needs it.

1. **The cell and its parts.** Plasma membrane, nucleus, mitochondria, endoplasmic reticulum. What each one is for.
   UNIT 1
2. **The four tissue types.** Epithelial, connective, muscle, nervous. Enough to tell them apart and say what each does.
   UNITS 1, 2, 5
3. **The neuron.** Cell body, dendrites, axon, axon terminal, myelin. Which end receives and which end sends.
   UNITS 2, 3
4. **How skeletal muscle is built.** Muscle, fiber, myofibril, sarcomere. Nested, biggest to smallest.
   UNIT 2
5. **The heart.** Four chambers, four valves, and the path blood takes through them in order.
   UNIT 4
6. **Blood vessel types.** Artery, arteriole, capillary, venule, vein, and which direction each carries blood.
   UNIT 4
7. **The respiratory tract.** Trachea down to alveolus, in order. What an alveolus is next to.
   UNIT 4
8. **The nephron.** Glomerulus, capsule, proximal tubule, loop, distal tubule, collecting duct. In order.
   UNIT 5
9. **The endocrine glands.** Where the pituitary, thyroid, adrenals and pancreas are. Not their hormones, just where they are.
   UNIT 3
10. **The digestive tract.** Mouth to anus in order, plus liver, gallbladder and pancreas hanging off it.
    UNIT 5

CHEMISTRY ASSUMED ON ENTRY

Not a chemistry course. These are the ideas physiology runs on, and each is revisited at the depth its unit requires rather than all at once.

11. **Ions and charge.** Sodium, potassium, calcium and chloride carry charge, and that charge is what makes a nerve fire and a muscle contract.
    UNITS 1, 2, 3, 4, 5
12. **Concentration and gradients.** What a concentration is, and what it means for something to be more concentrated on one side of a membrane than the other.
    UNITS 1, 2, 3, 4, 5
13. **Water, polarity and solubility.** Why water dissolves salt but not fat, and what hydrophilic and hydrophobic mean.
    UNITS 1, 3, 4, 5
14. **Acids, bases and pH.** What pH measures, which direction is more acidic, and roughly what blood pH is.
    UNITS 1, 4, 5
15. **Buffers.** What a buffer does when acid is added, in one sentence.
    UNITS 1, 5
16. **Proteins and shape.** That a protein does its job because of its shape, and that heat or a pH change can wreck that shape.
    UNITS 1, 2, 3
17. **ATP and energy.** That ATP is the cell's energy currency and that splitting it releases energy the cell can use.
    UNITS 1, 2, 5
18. **Units and simple calculation.** Reading and converting the units this course uses: molar and millimolar, mmHg, liters per minute, percent.
    UNITS 1, 2, 3, 4, 5

QUANTITATIVE SKILLS ASSUMED ON ENTRY

Arithmetic, units and graph reading. Nothing beyond it, but it has to be fluent, because a question about the kidney should not become a question about unit conversion.

19. **Metric prefixes and unit conversion.** Milli, micro and kilo, and moving between them without looking it up. Multiply molar by 1,000 to get millimolar.
    UNITS 1, 2, 3, 4, 5

20. **Counting particles, not molecules.** Osmolarity counts particles. A salt that splits into two ions contributes twice what a sugar that does not split contributes.
    UNITS 1, 2, 5

21. **Proportions and percent change.** If one number doubles and another halves, what happens to their product. Percent of a starting value, not of the final one.
    UNIT 4

22. **Rates, and what per means.** A rate is an amount per unit time, and it needs both parts. 5 liters is not a rate. 5 liters per minute is.
    UNITS 1, 4, 5

23. **Reading a graph.** Which variable goes on which axis, reading a value off a curve, and knowing that a steep part of a curve means a small change in x makes a big change in y.
    UNITS 1, 2, 4, 5

24. **What a log scale does.** You do not need to calculate logs. You need to know that one pH unit is a tenfold change, so 7.0 to 6.0 is ten times more acidic, not a little more.
    UNITS 1, 4, 5

25. **In minus out.** If more comes in than goes out the amount rises, and by how much. This is arithmetic, and it is the most used tool in the first unit.
    UNITS 1, 5

26. **Charge, and why mEq is not mM.** Milliequivalents count charge. For a singly charged ion the two numbers are the same. For calcium, which carries two charges, they are not.
    UNITS 1, 2

27. **Sanity checking an answer.** Knowing roughly what a normal value is, so an answer that is a hundred times off looks wrong to you before you hand it in.
    UNITS 1, 2, 3, 4, 5

Course competencies

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These 268 competencies are the whole of what BIO 005 asks you to be able to do. They are what the Mastery Physio OS tracks and they are what every exam is written from. Nothing is assessed that is not on this list.

How to read this list. Competencies are grouped first by unit, then by topic within the unit, in teaching order. Each one is tagged for where you are held to it: LECTURE means it is on the exam for that unit, and LAB means you produce it on real output, a trace, a curve, a set of values or a calculation. Many carry both, because in physiology explaining a mechanism and reading what it does are two different skills.

COMPETENCY LOAD BY UNIT

UNIT	TITLE	TOTAL	LECTURE	LAB	MUST KNOW
1	Foundations, Membranes and Cell Signaling	49	40	24	32
2	Neurophysiology and Muscle Physiology	50	44	29	34
3	Sensory, Motor, Autonomic and Endocrine Physiology	46	40	21	29
4	Cardiovascular and Respiratory Physiology	63	56	32	51
5	Renal, Digestive, Metabolic, Immune and Reproductive Physiology	60	54	26	46
	All five units	**268**	**234**	**132**	**192**

Unit 1. Foundations, Membranes and Cell Signaling

49 COMPETENCIES · WEEKS 1 TO 3 · MIDTERM 1

FOUNDATIONS OF PHYSIOLOGY

1. **Physiology and levels of function.** Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained. LECTURE
2. **Structure and function relationship.** Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems. LECTURE LAB
3. **Homeostasis defined.** Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state. LECTURE
4. **Feedback loop components.** Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step. LECTURE LAB
5. **Negative and positive feedback.** Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it. LECTURE
6. **Feedforward control and acclimatization.** Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario. LECTURE
7. **Local and reflex control pathways.** Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other. LECTURE
8. **Mass balance.** Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant. LECTURE LAB

9. **Body fluid compartments.** State the approximate volumes of total body water, intracellular fluid, extracellular fluid, plasma, and interstitial fluid in a 70 kg adult and compare the dominant solutes of the intracellular and extracellular compartments. LECTURE LAB
10. **Compartment separation and clinical volume shifts.** Predict the direction of water movement between compartments when extracellular osmolarity rises or falls and name a clinical situation that produces each shift. LECTURE

QUANTITATIVE SKILLS FOR PHYSIOLOGY

11. **Units and unit conversion.** Convert among the units used in physiology including molarity, osmolarity, milliequivalents, mmHg, liters per minute, and percent solutions. LECTURE LAB
12. **Graphing and data interpretation.** Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set. LAB
13. **Experimental design and controls.** Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out. LAB
14. **Measurement error and variability.** Distinguish random from systematic error and explain why physiological measurements are repeated and averaged. LAB

CHEMICAL FOUNDATIONS

15. **Water and solution properties.** Explain how the polarity and hydrogen bonding of water determine solubility and identify whether a given solute is hydrophilic or hydrophobic. LECTURE
16. **pH and buffers.** Define pH, state the normal pH range of arterial blood, and explain how a buffer pair resists a change in pH when acid or base is added. LECTURE LAB
17. **Protein structure and function.** Relate the levels of protein structure to binding site shape and explain how denaturation by heat or pH change destroys function. LECTURE LAB
18. **Enzyme activity and regulation.** Describe how enzymes lower activation energy and predict the effect of substrate concentration, temperature, pH, and competitive or allosteric inhibition on reaction rate. LECTURE LAB
19. **ATP and energy coupling.** Explain how ATP hydrolysis is coupled to endergonic cellular work and name three categories of work that require it. LECTURE
20. **Enzyme assay.** Measure enzyme activity across a range of temperature or pH using a spectrophotometric or colorimetric assay, plot the results, and identify the optimum. LAB

MEMBRANE STRUCTURE AND DIFFUSION

21. **Membrane composition and fluidity.** Describe the fluid mosaic membrane and state how phospholipids, cholesterol, glycolipids, and integral and peripheral proteins contribute to its properties. LECTURE
22. **Determinants of permeability.** Rank molecules by their ability to cross a lipid bilayer unaided using size, charge, and lipid solubility, and predict which will require a protein. LECTURE
23. **Simple diffusion and Fick's law.** State the variables in Fick's law of diffusion and predict how a change in concentration gradient, surface area, membrane thickness, or distance changes the rate of transfer. LECTURE LAB
24. **Osmolarity and tonicity.** Calculate the osmolarity of a solution, distinguish osmolarity from tonicity, and classify a solution as isotonic, hypotonic, or hypertonic to a cell. LECTURE LAB
25. **Osmosis and cell volume.** Predict the direction of water movement and the resulting change in cell volume when a cell is placed in a solution of stated osmolarity and penetrating solute content. LECTURE LAB
26. **Diffusion and osmosis experiment.** Measure diffusion and osmotic movement across a selectively permeable membrane and relate the observed rate to molecular size and concentration gradient. LAB
27. **Tonicity and red blood cells.** Observe erythrocytes in solutions of different tonicity, identify crenation, normal shape, and hemolysis, and explain each result. LAB

MEMBRANE TRANSPORT

28. **Facilitated diffusion.** Describe carrier and channel mediated diffusion and explain why carrier mediated transport shows saturation and specificity while simple diffusion does not. LECTURE
29. **Primary active transport.** Explain how the sodium potassium ATPase uses ATP to move three sodium out and two potassium in, and state the two gradients it maintains. LECTURE LAB
30. **Secondary active transport.** Distinguish symport from antiport and trace how the sodium gradient powers glucose uptake by SGLT and calcium removal by the sodium calcium exchanger. LECTURE

31. **Transport maximum and saturation.** Interpret a transport rate curve, identify the transport maximum, and apply the concept to renal glucose handling in hyperglycemia. LECTURE LAB

32. **Vesicular transport.** Compare phagocytosis, pinocytosis, receptor mediated endocytosis, and exocytosis by trigger, cargo, and energy requirement. LECTURE

33. **Transepithelial transport.** Trace glucose or sodium from lumen to blood across a polarized epithelium and identify which step occurs at the apical membrane and which at the basolateral membrane. LECTURE

34. **Transport simulation.** Use a membrane transport simulation to distinguish simple diffusion from facilitated diffusion and active transport by their response to gradient reversal and metabolic poison. LAB

MEMBRANE POTENTIAL

35. **Ion distribution and electrochemical gradients.** State the typical intracellular and extracellular concentrations of sodium, potassium, chloride, and calcium and separate the chemical from the electrical component of the driving force on each ion. LECTURE

36. **Nernst equation.** Calculate the equilibrium potential for an ion with the Nernst equation and explain what the sign of the result means for the direction that ion will move. LECTURE LAB

37. **Resting membrane potential.** Explain why the resting membrane potential sits near the potassium equilibrium potential and predict how it shifts when membrane permeability to potassium or sodium changes. LECTURE

38. **Ion channel gating.** Compare leak, voltage gated, ligand gated, and mechanically gated channels by what opens each one and give a physiological location for each. LECTURE

39. **Depolarization and hyperpolarization.** Define depolarization, repolarization, hyperpolarization, and overshoot and label each on a membrane potential tracing. LECTURE LAB

40. **Membrane potential simulation.** Manipulate extracellular potassium and sodium in a simulation and record the resulting change in resting membrane potential against the Nernst prediction. LAB

CELL SIGNALING

41. **Signal types and range.** Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel. LECTURE

42. **Receptor location and ligand solubility.** Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response. LECTURE

43. **G protein coupled receptors.** Trace a G protein coupled receptor pathway from ligand binding through the G protein and amplifier enzyme to the second messenger and the cellular response. LECTURE

44. **Second messengers.** Identify cyclic AMP, IP3, diacylglycerol, and calcium as second messengers and state the enzyme that generates each and one target it activates. LECTURE

45. **Catalytic receptors and intracellular receptors.** Compare a receptor enzyme such as the insulin receptor with an intracellular steroid receptor by mechanism and by time course of the response. LECTURE

46. **Signal amplification.** Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway. LECTURE

47. **Receptor modulation.** Define agonist, antagonist, competitive inhibition, up regulation, and down regulation and predict the effect of chronic ligand excess on target cell sensitivity. LECTURE

48. **Signal termination.** Name three mechanisms that end a chemical signal and explain why a signal that cannot be terminated produces pathology. LECTURE

49. **Dose response relationships.** Plot a dose response curve, identify threshold, maximal response, and EC50, and compare a full agonist with a partial agonist on the same axes. LAB

Unit 2. Neurophysiology and Muscle Physiology

50 COMPETENCIES · WEEKS 4 TO 6 · MIDTERM 1

NEURONS AND NEUROGLIA

50. **Neuron structural and functional classes.** Classify a neuron as multipolar, bipolar, or pseudounipolar by structure and as sensory, motor, or interneuron by function, and match each class to a location in the nervous system. LECTURE LAB

51. **Functional regions of a neuron.** Label the dendrites, cell body, axon hillock, trigger zone, axon, and axon terminal and state which signal type each region carries. LECTURE LAB

52. **Glial cell functions.** Match astrocytes, oligodendrocytes, microglia, ependymal cells, satellite cells, and Schwann cells to their physiological roles. LECTURE LAB

53. **Myelin and its loss.** Explain how myelin and the nodes of Ranvier speed conduction and predict the functional consequence of demyelination. LECTURE

54. **Axonal transport.** Distinguish fast anterograde and retrograde axonal transport from slow transport and state what each carries. LECTURE

ELECTRICAL SIGNALING

55. **Graded potentials.** Describe how a graded potential is produced and explain why it varies with stimulus strength and decays with distance, using the terms current leak and cytoplasmic resistance. LECTURE

56. **Action potential phases.** Diagram an action potential and state the channel state and ion movement responsible for depolarization to peak and for repolarization and afterhyperpolarization. LECTURE LAB

57. **Threshold and all or none.** Explain what threshold represents at the trigger zone and why action potential amplitude does not change with a stronger stimulus. LECTURE

58. **Coding of stimulus intensity.** Explain how the nervous system encodes stimulus strength when action potential amplitude is fixed. LECTURE

59. **Refractory periods.** Distinguish the absolute from the relative refractory period by channel state and explain how the refractory period sets maximum firing frequency and prevents backward conduction. LECTURE

60. **Conduction velocity.** Rank axons by conduction velocity using diameter and myelination and contrast continuous with saltatory conduction. LECTURE LAB

61. **Effects of altered extracellular ions.** Predict the effect of hyperkalemia and hypokalemia on resting potential and excitability and explain why local anesthetics that block sodium channels abolish the action potential. LECTURE

62. **Action potential simulation.** Run a neuron simulation with sodium and potassium channel blockers and interpret the resulting change in the action potential tracing. LAB

63. **Nerve conduction measurement.** Measure conduction velocity from a recorded compound action potential and account for the difference between the fastest and slowest fibers. LAB

SYNAPTIC TRANSMISSION

64. **Sequence at a chemical synapse.** Order the events of chemical synaptic transmission from action potential arrival through calcium entry and vesicle fusion to receptor binding and postsynaptic response. LECTURE LAB

65. **Neurotransmitter classes.** Match acetylcholine, the catecholamines, serotonin, glutamate, GABA, glycine, and the neuropeptides to their usual excitatory or inhibitory effect and to a site of action. LECTURE

66. **Neurotransmitter removal.** Name the three routes that clear a neurotransmitter from the synaptic cleft, degradation, reuptake, and diffusion, and give a drug that blocks one of them. LECTURE

67. **Excitatory and inhibitory postsynaptic potentials.** Distinguish an EPSP from an IPSP by the ion channel opened and the direction of the membrane potential change. LECTURE LAB

68. **Summation and integration.** Distinguish temporal from spatial summation and determine whether a stated combination of EPSPs and IPSPs will bring the trigger zone to threshold. LECTURE LAB

69. **Presynaptic modulation.** Compare presynaptic inhibition and facilitation with postsynaptic modulation and state the advantage of presynaptic control at a single input, which is selectivity. LECTURE

70. **Synaptic plasticity.** Explain long term potentiation as a mechanism of learning and identify the roles of repeated stimulation and receptor insertion. LECTURE

71.

Electrical synapses. Compare an electrical synapse at a gap junction with a chemical synapse by speed and by capacity for modulation. LECTURE

72. **Synapse simulation.** Alter neurotransmitter release and receptor availability in a synapse simulation and interpret the resulting postsynaptic recording. LAB

CENTRAL INTEGRATION AND REFLEXES

73. **Reflex arc components.** Diagram a reflex arc naming the receptor, sensory neuron, integrating center, motor neuron, and effector, and classify the reflex as monosynaptic or polysynaptic. LECTURE LAB
74. **Muscle spindle and stretch reflex.** Explain how the muscle spindle detects stretch and trace the stretch reflex including reciprocal inhibition of the antagonist. LECTURE LAB
75. **Golgi tendon organ.** Contrast the stimulus and the reflex response of the Golgi tendon organ with those of the muscle spindle. LECTURE
76. **Withdrawal and crossed extensor reflexes.** Trace the flexor withdrawal reflex and the crossed extensor reflex and explain how the two together preserve balance. LECTURE LAB
77. **Ascending and descending pathways.** Compare the dorsal column, spinothalamic, and corticospinal pathways by the information carried, the side of decussation, and the deficit produced by a lesion. LECTURE
78. **Cerebrospinal fluid and the blood brain barrier.** State the functions of cerebrospinal fluid and explain how the blood brain barrier selects what reaches neurons, including why lipid soluble drugs cross readily. LECTURE
79. **Reflex testing and reaction time.** Elicit and grade deep tendon reflexes and measure reaction time to distinguish a reflex response from a voluntary response by latency. LAB

SKELETAL MUSCLE PHYSIOLOGY

80. **Neuromuscular junction.** Trace transmission at the neuromuscular junction from the motor neuron action potential to the end plate potential and explain why every action potential in the motor neuron produces one in the muscle fiber. LECTURE LAB
81. **Excitation contraction coupling.** Trace excitation contraction coupling from the T tubule through the DHP and ryanodine receptors to calcium release from the sarcoplasmic reticulum. LECTURE LAB
82. **Crossbridge cycle.** Order the steps of the crossbridge cycle, binding, power stroke, detachment, and reactivation, and state where ATP binds and where it is hydrolyzed. LECTURE LAB
83. **Calcium and the regulatory proteins.** Explain how calcium binding to troponin moves tropomyosin off the myosin binding site and predict what happens to contraction if calcium cannot be released. LECTURE
84. **Relaxation and calcium removal.** Explain how SERCA and acetylcholinesterase end a contraction and predict the result if either one fails. LECTURE
85. **The muscle twitch.** Label the latent period, contraction phase, and relaxation phase on a twitch tracing and explain what limits each phase. LECTURE LAB
86. **Summation and tetanus.** Explain wave summation and distinguish unfused from fused tetanus by stimulus frequency and calcium accumulation. LECTURE LAB
87. **Motor units and recruitment.** Define the motor unit and explain how size of the unit and orderly recruitment grade the force a whole muscle produces. LECTURE LAB
88. **Length tension relationship.** Interpret a length tension curve and explain why tension falls at very short and very long sarcomere lengths. LECTURE LAB
89. **Isometric and isotonic contraction.** Distinguish isometric, concentric, and eccentric contractions and predict shortening velocity from the load applied. LECTURE LAB
90. **Muscle fiber types.** Compare slow oxidative, fast oxidative glycolytic, and fast glycolytic fibers by speed, ATP source, fatigue resistance, and typical task. LECTURE LAB
91. **ATP sources during activity.** Order creatine phosphate, anaerobic glycolysis, and oxidative phosphorylation by how quickly each supplies ATP and how long each can sustain it. LECTURE
92. **Fatigue and oxygen debt.** Distinguish central from peripheral fatigue and explain the metabolic basis of excess post exercise oxygen consumption. LECTURE LAB
93. **Muscle plasticity.** Predict the change in fiber size, capillary supply, and mitochondrial content produced by endurance training, resistance training, and disuse. LECTURE
94. **Electromyography and grip fatigue.** Record grip force and surface EMG over a sustained contraction and relate the decline in force to recruitment and fatigue. LAB

95. **Twitch and tetanus simulation.** Generate twitch and tetanus tracings in a muscle simulation by varying stimulus voltage and frequency and identify threshold and maximal stimulus. LAB

CARDIAC AND SMOOTH MUSCLE

96. **Smooth muscle contraction mechanism.** Trace smooth muscle contraction through calcium entry and release to calmodulin and myosin light chain kinase and explain why the mechanism is slower and more economical than in skeletal muscle. LECTURE

97. **Smooth muscle types and regulation.** Compare single unit and multi unit smooth muscle and list the stimuli that alter smooth muscle tone including stretch and local chemical signals. LECTURE

98. **Cardiac muscle properties.** Explain how intercalated discs and gap junctions allow the myocardium to act as a functional syncytium and state why cardiac muscle cannot be tetanized. LECTURE LAB

99. **Comparison of the three muscle types.** Build a comparison of skeletal, cardiac, and smooth muscle by calcium source, regulatory protein, control, and speed of contraction. LECTURE LAB

Unit 3. Sensory, Motor, Autonomic and Endocrine Physiology

46 COMPETENCIES · WEEKS 7 TO 9 · MIDTERMS 1 AND 2

GENERAL SENSORY PHYSIOLOGY

100. **Sensory transduction.** Explain how a stimulus is converted into a receptor potential and how that graded potential becomes a train of action potentials in the sensory neuron. LECTURE
101. **Receptor classification.** Classify a receptor by stimulus type as a mechanoreceptor, chemoreceptor, thermoreceptor, photoreceptor, or nociceptor and give a body location for each. LECTURE LAB
102. **Stimulus coding.** Explain how the nervous system encodes modality, location, intensity, and duration and identify the labeled line principle. LECTURE
103. **Receptive fields and acuity.** Relate receptive field size and lateral inhibition to two point discrimination and predict which body regions have the finest acuity. LECTURE LAB
104. **Receptor adaptation.** Distinguish tonic from phasic receptors by their adaptation rate and match each to a stimulus the body needs to monitor continuously or only at onset. LECTURE LAB
105. **Somatosensory pathways.** Trace touch and pain information from receptor to cortex and explain why the somatosensory homunculus is disproportionate. LECTURE
106. **Pain and its modulation.** Distinguish fast and slow pain by fiber type, explain referred pain by convergence, and describe how gate control and endogenous opioids reduce pain transmission. LECTURE
107. **Tactile mapping and adaptation testing.** Measure two point discrimination at several body sites and show receptor adaptation and referred sensation experimentally. LAB

SPECIAL SENSES

108. **Optics and accommodation.** Explain refraction by the cornea and lens and describe how accommodation and the pupillary reflex focus near and far images. LECTURE LAB
109. **Phototransduction.** Trace phototransduction from photon absorption by retinal through the change in cyclic GMP to the decrease in glutamate release and explain why photoreceptors hyperpolarize to light. LECTURE
110. **Rods cones and visual processing.** Compare rods and cones by sensitivity and acuity and trace the visual pathway to predict the field deficit produced by a lesion at the optic nerve or the optic chiasm. LECTURE
111. **Refractive errors and clinical vision.** Explain myopia, hyperopia, presbyopia, and astigmatism in terms of focal point and eyeball or lens shape and state the corrective lens for each. LECTURE LAB
112. **Auditory transduction.** Trace sound from the tympanic membrane through the ossicles and cochlear fluid to hair cell bending and auditory nerve firing and explain how pitch and loudness are coded. LECTURE
113. **Conductive and sensorineural hearing loss.** Distinguish conductive from sensorineural hearing loss by the site of the lesion and by the expected Weber and Rinne results. LECTURE LAB
114. **Equilibrium.** Explain how the semicircular canals detect angular acceleration and how the utricle and saccule detect linear acceleration and head position. LECTURE LAB
115. **Taste and smell.** Compare gustatory and olfactory transduction and explain why olfactory input reaches the limbic system without a thalamic relay. LECTURE
116. **Vision testing.** Perform visual acuity, blind spot, accommodation, and color vision testing and interpret each result. LAB
117. **Hearing and equilibrium testing.** Perform Weber and Rinne tuning fork tests and a balance test and interpret the findings. LAB

MOTOR CONTROL

118. **Motor control hierarchy.** Order the levels of motor control from spinal reflex through brainstem to cortex and state what each level contributes to a voluntary movement. LECTURE
119. **Corticospinal pathway.** Trace the corticospinal pathway from the primary motor cortex to the skeletal muscle and identify where it crosses. LECTURE
120. **Cerebellum and basal ganglia.** Contrast the contributions of the cerebellum and the basal ganglia to movement and match a described motor sign to the structure involved. LECTURE

121. **Upper and lower motor neuron signs.** Distinguish upper from lower motor neuron lesions by tone, reflexes, and muscle bulk and assign a set of findings to the correct level. LECTURE LAB

AUTONOMIC NERVOUS SYSTEM

122. **Autonomic and somatic organization.** Contrast the somatic motor system with the autonomic two neuron pathway by neuron count and by effector tissue. LECTURE

123. **Sympathetic and parasympathetic divisions.** Compare the two autonomic divisions by CNS origin and ganglion location and predict the effect of each on heart rate and on the gastrointestinal tract. LECTURE LAB

124. **Autonomic neurotransmitters and receptors.** Match nicotinic, muscarinic, alpha, and beta receptors to their neurotransmitter, their location, and the response each produces. LECTURE

125. **Dual innervation and tonic control.** Explain dual innervation and autonomic tone and predict the effect of removing one division from a dually innervated organ. LECTURE

126. **Adrenal medulla.** Explain why the adrenal medulla is a modified sympathetic ganglion and compare the duration of its circulating catecholamine effect with direct sympathetic innervation. LECTURE

127. **Autonomic pharmacology.** Predict the effect of a beta blocker or a muscarinic antagonist on a named organ from the receptor it targets. LECTURE

128. **Autonomic function testing.** Measure heart rate variability and the response to a cold pressor or orthostatic challenge and interpret the result as sympathetic or parasympathetic dominance. LAB

ENDOCRINE PRINCIPLES

129. **Hormone classes.** Classify hormones as peptide, steroid, or amine and compare the three by synthesis, storage, and receptor location. LECTURE

130. **Hormone transport and half life.** Explain how protein binding in plasma affects the free hormone fraction and the half life and predict which hormone class circulates bound. LECTURE

131. **Hormone receptors and target response.** Explain why one hormone can produce different responses in different tissues and relate receptor number to target cell sensitivity. LECTURE

132. **Hormone interactions.** Define synergism, permissiveness, and antagonism and identify each in a described hormone pair. LECTURE

133. **Control of hormone release.** Classify a hormone stimulus as humoral, neural, or hormonal and trace an example of each. LECTURE

134. **Hypothalamic pituitary axes.** Trace a three tier axis from hypothalamic releasing hormone through the anterior pituitary trophic hormone to the peripheral gland and place the long and short loop negative feedback. LECTURE LAB

135. **Primary and secondary endocrine disorders.** Use hormone levels at two tiers of an axis to classify a disorder as primary or secondary and as hypersecretion or hyposecretion. LECTURE LAB

ENDOCRINE GLANDS

136. **Posterior pituitary hormones.** Explain why the posterior pituitary is neural tissue and state the stimulus and target action of antidiuretic hormone and oxytocin. LECTURE

137. **Growth hormone.** Describe the direct and insulin like growth factor mediated actions of growth hormone and predict the result of excess or deficiency before and after epiphyseal closure. LECTURE

138. **Thyroid hormone.** Trace thyroid hormone synthesis, state the metabolic actions of T3 and T4, and match hyperthyroid and hypothyroid signs to the underlying rate of metabolism. LECTURE LAB

139. **Adrenal cortex.** Match the three cortical zones to aldosterone, cortisol, and the adrenal androgens and state the primary action of each. LECTURE LAB

140. **Cortisol and the stress response.** Describe the metabolic, immune, and cardiovascular actions of cortisol and explain the consequences of chronic elevation. LECTURE

141. **Calcium homeostasis.** Trace the response to falling plasma calcium through parathyroid hormone, calcitriol, and the actions on bone, kidney, and intestine, and state the role of calcitonin. LECTURE LAB

142. **Pancreatic islet hormones.** Compare insulin and glucagon by stimulus and by effect on liver glycogen and on plasma glucose and identify the islet cell that secretes each. LECTURE LAB

143. **Diabetes mellitus.** Distinguish type 1 from type 2 diabetes by mechanism and explain how untreated hyperglycemia produces polyuria and ketoacidosis. LECTURE

144. **Glucose tolerance testing.** Plot a glucose tolerance curve from measured or simulated data and distinguish a normal response from an impaired one. LAB

145.

Hormone assay. Run or simulate an immunoassay for a hormone and use the standard curve to determine an unknown concentration. LAB

Unit 4. Cardiovascular and Respiratory Physiology

63 COMPETENCIES · WEEKS 10 TO 12 · MIDTERM 2

BLOOD

146. **Plasma and blood functions.** State the three functional categories of blood, transport, regulation, and protection, and name the major plasma proteins and the function of each. LECTURE LAB

147. **Hematocrit and formed elements.** Define hematocrit and interpret a value as normal or abnormal in the context of anemia and dehydration. LECTURE LAB

148. **Erythrocyte structure and hemoglobin.** Relate the biconcave shape and absent nucleus of the erythrocyte to its function and describe how the four heme groups of hemoglobin carry oxygen. LECTURE LAB

149. **Erythropoiesis and its regulation.** Trace the erythropoietin negative feedback loop from tissue hypoxia to increased red cell production and state the nutrients required. LECTURE

150. **Red cell destruction and bilirubin.** Trace the fate of hemoglobin after red cell breakdown and explain how failure to clear bilirubin produces jaundice. LECTURE

151. **Leukocytes.** Match neutrophils, lymphocytes, monocytes, eosinophils, and basophils to their defensive function and to the condition that raises each. LECTURE LAB

152. **Hemostasis.** Order the three phases of hemostasis, vascular spasm, platelet plug formation, and coagulation, and identify the role of von Willebrand factor and thromboxane. LECTURE

153. **Coagulation cascade.** Compare the intrinsic and extrinsic pathways to the common pathway and identify the roles of thrombin, fibrin, calcium, and vitamin K. LECTURE

154. **Clot limitation and fibrinolysis.** Explain how anticoagulants and plasmin limit and remove a clot and predict the effect of a drug that blocks vitamin K or platelet aggregation. LECTURE

155. **ABO and Rh blood types.** Predict agglutination for any donor and recipient ABO and Rh combination and explain the mechanism of hemolytic disease of the newborn. LECTURE LAB

156. **Hematocrit and hemoglobin determination.** Determine hematocrit and hemoglobin from a sample or simulation and evaluate the result against reference ranges. LAB

157. **Blood typing.** Perform simulated ABO and Rh typing and determine compatible donor types for the result obtained. LAB

CARDIAC ELECTROPHYSIOLOGY

158. **Pacemaker potential.** Explain the unstable pacemaker potential of the sinoatrial node by the funny current, calcium entry, and potassium exit and state what sets intrinsic heart rate. LECTURE

159. **Contractile cell action potential.** Diagram the ventricular action potential and identify the ion movement responsible for the rapid upstroke, the plateau, and repolarization. LECTURE LAB

160. **Conduction system.** Trace an impulse through the sinoatrial node, internodal pathways, atrioventricular node, bundle of His, bundle branches, and Purkinje fibers and explain the purpose of the atrioventricular delay. LECTURE LAB

161. **Cardiac refractory period.** Explain why the long refractory period of cardiac muscle prevents tetanus and why that matters for pump function. LECTURE

162. **ECG waves and intervals.** Label the P wave, QRS complex, T wave, PR interval, and QT interval and state the electrical event each represents. LECTURE LAB

163. **ECG interpretation.** Calculate heart rate from an ECG strip and identify sinus rhythm, tachycardia, bradycardia, and a first degree or complete heart block. LECTURE LAB

164. **ECG recording.** Record a lead II ECG, measure the intervals, and correlate each wave with the mechanical event that follows it. LAB

CARDIAC MECHANICS

165. **Cardiac cycle.** Order the phases of the cardiac cycle and state the pressure relationship that opens and closes the atrioventricular and semilunar valves in each phase. LECTURE LAB

166. **Pressure volume loop.** Label the four phases of a ventricular pressure volume loop and predict how the loop changes with increased preload, increased afterload, or increased contractility. LECTURE

167. **Heart sounds.** Explain the origin of the first and second heart sounds and relate a murmur to valve stenosis or regurgitation by its timing. LECTURE LAB

168. **Stroke volume and ejection fraction.** Calculate stroke volume and ejection fraction from end diastolic and end systolic volumes and interpret a reduced ejection fraction. LECTURE LAB

169. **Cardiac output.** Calculate cardiac output and cardiac reserve and predict the effect of a change in heart rate or stroke volume on each. LECTURE LAB

170. **Frank Starling relationship.** State the Frank Starling law and explain how venous return and end diastolic volume determine the force of the next contraction. LECTURE

171. **Preload afterload and contractility.** Distinguish preload, afterload, and contractility and predict how a change in each alters stroke volume. LECTURE

172. **Autonomic regulation of the heart.** Explain how sympathetic and parasympathetic input alter heart rate and contractility at the receptor and second messenger level. LECTURE

173. **Heart sound and pulse correlation.** Auscultate the heart sounds, locate the valve areas, and correlate the sounds with the pulse and with the cardiac cycle phases. LAB

VASCULAR PHYSIOLOGY

174. **Pressure flow and resistance.** Apply the relationship among flow, pressure gradient, and resistance and predict the effect of a change in vessel radius, blood viscosity, or vessel length on flow. LECTURE LAB

175. **Vessel structure and function.** Match arteries, arterioles, capillaries, venules, and veins to their role in pressure buffering, resistance, exchange, and capacitance. LECTURE LAB

176. **Arterial blood pressure.** Define systolic, diastolic, pulse, and mean arterial pressure, calculate mean arterial pressure, and explain how it is determined by cardiac output and total peripheral resistance. LECTURE LAB

177. **Local control of blood flow.** Explain myogenic, metabolic, and endothelial control of arteriolar tone and predict the local flow response to increased tissue metabolism. LECTURE

178. **Capillary exchange and Starling forces.** Identify the four Starling forces at a capillary and determine the net direction of fluid movement at the arterial and venous ends. LECTURE

179. **Edema.** Explain how increased capillary hydrostatic pressure, reduced plasma protein, increased permeability, or lymphatic blockage each produce edema and match a clinical scenario to its mechanism. LECTURE

180. **Lymphatic return.** State the volume of fluid the lymphatic system returns each day and trace lymph from an interstitial space to the subclavian vein. LECTURE

181. **Venous return.** Explain the skeletal muscle pump, the respiratory pump, and venoconstriction and predict the effect of prolonged standing on venous return and stroke volume. LECTURE LAB

182. **Blood pressure measurement.** Measure blood pressure by auscultation, explain the origin of the Korotkoff sounds, and record the response to a postural or exercise challenge. LAB

CARDIOVASCULAR REGULATION

183. **Baroreceptor reflex.** Trace the baroreceptor reflex from the carotid sinus and aortic arch through the medulla to the heart and vessels and state the response to a fall in blood pressure. LECTURE LAB

184. **Hormonal control of blood pressure.** Describe the roles of angiotensin II, aldosterone, antidiuretic hormone, and atrial natriuretic peptide in long term blood pressure control. LECTURE

185. **Cardiovascular response to exercise.** Predict the change in heart rate, stroke volume, cardiac output, total peripheral resistance, and regional blood flow distribution during moderate exercise. LECTURE LAB

186. **Compensation in hemorrhage and shock.** Sequence the compensatory responses to blood loss and explain the point at which compensation fails. LECTURE

RESPIRATORY MECHANICS

187. **Functions and functional zones.** State the functions of the respiratory system and distinguish the conducting zone from the respiratory zone by structure and by role in gas exchange. LECTURE LAB

188. **Pressure gradients and airflow.** Apply Boyle's law to inspiration and expiration and relate alveolar pressure to atmospheric pressure at each point in a quiet breath. LECTURE

189. **Intrapleural pressure.** Explain why intrapleural pressure is subatmospheric and predict what happens to the lung and chest wall when the pleural seal is broken. LECTURE

190. **Compliance and elastic recoil.** Define lung compliance and elastic recoil and predict the effect of fibrosis and of emphysema on each. LECTURE

191. **Surfactant and surface tension.** Explain how surfactant lowers surface tension and why its absence causes alveolar collapse in the premature newborn. LECTURE

192. **Airway resistance.** Identify the main determinant of airway resistance and predict the effect of bronchoconstriction, mucus, and sympathetic stimulation on airflow. LECTURE

193. **Lung volumes and capacities.** Define tidal volume, inspiratory and expiratory reserve volume, residual volume, vital capacity, and total lung capacity and read each from a spirogram. LECTURE LAB

194. **Obstructive and restrictive patterns.** Use the ratio of forced expiratory volume in one second to forced vital capacity to classify a spirometry result as obstructive or restrictive. LECTURE LAB

195. **Alveolar ventilation and dead space.** Calculate minute ventilation and alveolar ventilation and explain why slow deep breathing ventilates the alveoli better than rapid shallow breathing. LECTURE LAB

196. **Spirometry.** Record a spirogram, calculate the lung volumes and capacities available from it, and compare the results with predicted values. LAB

GAS EXCHANGE AND TRANSPORT

197. **Partial pressures.** Apply Dalton's law to calculate the partial pressure of oxygen in inspired and alveolar air and state the normal partial pressures in alveolar air, arterial blood, and venous blood. LECTURE

198. **Diffusion at the respiratory membrane.** Apply the determinants of diffusion to the respiratory membrane and predict the effect of edema, fibrosis, and emphysema on gas transfer. LECTURE

199. **Ventilation perfusion matching.** Explain how local hypoxic vasoconstriction and bronchiolar responses match ventilation to perfusion and predict the consequence of a mismatch. LECTURE

200. **Oxygen transport and the oxyhemoglobin curve.** Explain the sigmoid shape of the oxyhemoglobin dissociation curve and state the physiological advantage of the steep and the flat portions. LECTURE LAB

201. **Curve shifts.** Predict the direction of the oxyhemoglobin curve shift produced by a change in pH, carbon dioxide, temperature, or 2,3 bisphosphoglycerate and state what the shift means for tissue oxygen delivery. LECTURE LAB

202. **Carbon dioxide transport.** Name the three forms in which carbon dioxide is carried, state the approximate percentage of each, and trace the chloride shift in a systemic capillary. LECTURE

203. **Bohr and Haldane effects.** Distinguish the Bohr effect from the Haldane effect and explain how the two work together at the tissue and at the lung. LECTURE

204. **Oxygen content versus partial pressure.** Distinguish oxygen content from oxygen partial pressure and saturation and explain why anemia and carbon monoxide poisoning reduce delivery differently. LECTURE

CONTROL OF VENTILATION

205. **Respiratory centers.** Identify the medullary and pontine respiratory centers and explain how the basic breathing rhythm is generated and modified. LECTURE

206. **Chemoreceptor control.** Compare central and peripheral chemoreceptors by stimulus and explain why carbon dioxide is the primary minute to minute drive to breathe. LECTURE

207. **Ventilation in exercise and at altitude.** Predict the ventilatory response to exercise and to acute and chronic altitude exposure and explain the acclimatization changes. LECTURE

208. **Ventilatory response testing.** Measure the change in breathing rate and depth after hyperventilation, breath holding, and rebreathing and explain each result by the change in carbon dioxide. LAB

Unit 5. Renal, Digestive, Metabolic, Immune and Reproductive Physiology

60 COMPETENCIES · WEEKS 13 TO 15 · MIDTERM 2

RENAL PHYSIOLOGY

209. **Kidney functions.** List the homeostatic functions of the kidney including water and electrolyte balance, acid base regulation, waste excretion, blood pressure control, and hormone production. LECTURE

210. **Nephron structure and function.** Match each nephron segment to its dominant transport function and distinguish a cortical from a juxtamedullary nephron by capability. LECTURE LAB

211. **Three renal processes.** Define filtration, reabsorption, secretion, and excretion and write the equation that relates them for any solute. LECTURE

212. **Filtration membrane and selectivity.** Describe the three layers of the filtration membrane and predict which plasma components appear in the filtrate based on size and charge. LECTURE LAB

213. **Net filtration pressure and GFR.** Calculate net filtration pressure from glomerular hydrostatic pressure, capsular pressure, and colloid osmotic pressure and predict the effect of each change on glomerular filtration rate. LECTURE LAB

214. **Regulation of GFR.** Compare myogenic autoregulation, tubuloglomerular feedback, sympathetic control, and hormonal control and predict the effect of afferent and efferent arteriolar constriction on filtration rate. LECTURE

215. **Renal clearance.** Calculate the clearance of a substance and use inulin and creatinine clearance to estimate glomerular filtration rate. LECTURE LAB

216. **Clearance and handling inference.** Use a clearance value relative to glomerular filtration rate to determine whether a substance is net reabsorbed or net secreted. LECTURE LAB

217. **Proximal tubule reabsorption.** Explain how the sodium gradient drives reabsorption of glucose, amino acids, and water in the proximal tubule and state why reabsorption there is described as isosmotic. LECTURE

218. **Transport maximum and renal threshold.** Distinguish transport maximum from renal threshold and explain the appearance of glucose in urine when plasma glucose is elevated. LECTURE LAB

219. **Tubular secretion.** State the purpose of tubular secretion and name substances secreted including hydrogen ion, potassium, and organic anions such as drugs. LECTURE

220. **Countercurrent mechanism.** Explain how the countercurrent multiplier of the loop of Henle and the countercurrent exchanger of the vasa recta build and preserve the medullary osmotic gradient. LECTURE

221. **Antidiuretic hormone and urine concentration.** Trace the response to dehydration from osmoreceptor firing through antidiuretic hormone release to aquaporin insertion and concentrated urine. LECTURE LAB

222. **Renin angiotensin aldosterone system.** Trace the renin angiotensin aldosterone pathway from the stimulus for renin release to the effects of angiotensin II and aldosterone on vessels, tubules, and blood pressure. LECTURE LAB

223. **Natriuretic peptides.** Explain how atrial natriuretic peptide opposes the renin angiotensin aldosterone system and state the stimulus for its release. LECTURE

224. **Potassium handling.** Explain how aldosterone and plasma potassium regulate potassium secretion in the collecting duct and predict the effect of a potassium wasting diuretic. LECTURE

225. **Micturition.** Trace the micturition reflex and explain how the internal and external sphincters allow voluntary control. LECTURE

226. **Urinalysis.** Perform a urinalysis measuring specific gravity, pH, glucose, protein, ketones, and blood and relate any abnormal finding to the renal process that failed. LAB

227. **Renal function calculation.** Calculate glomerular filtration rate, clearance, and filtered load from a data set and interpret the values in a clinical case. LAB

ACID BASE AND FLUID BALANCE

228. **Buffer systems.** Compare the bicarbonate, phosphate, and protein buffer systems by location and speed and explain why bicarbonate is the dominant extracellular buffer. LECTURE

229. **Respiratory control of pH.** Explain how a change in ventilation shifts the carbon dioxide and bicarbonate equilibrium and therefore plasma pH. LECTURE

230. **Renal control of pH.** Describe how the kidney secretes hydrogen ion and reclaims or generates bicarbonate and explain why renal compensation is slower and more complete than respiratory compensation. LECTURE

231. **The four acid base disorders.** Classify a disorder as respiratory or metabolic and as acidosis or alkalosis from pH, partial pressure of carbon dioxide, and bicarbonate. LECTURE LAB

232. **Compensation.** Determine whether an acid base disorder is uncompensated, partially compensated, or fully compensated and name the system doing the compensating. LECTURE LAB

233. **Arterial blood gas interpretation.** Interpret a set of arterial blood gas values and match the result to a plausible clinical cause. LAB

234. **Volume and osmolarity disturbances.** Classify a disturbance as volume depletion or overload and as hypoosmotic, isosmotic, or hyperosmotic and predict the resulting fluid shift between compartments. LECTURE

DIGESTIVE PHYSIOLOGY

235. **Four digestive processes.** Distinguish motility, secretion, digestion, and absorption and identify where in the tract each dominates. LECTURE

236. **Motility patterns.** Compare peristalsis, segmentation, and the migrating motor complex by pattern and purpose. LECTURE LAB

237. **Neural and hormonal regulation.** Explain short and long reflex control by the enteric nervous system and match gastrin, secretin, cholecystokinin, and glucose dependent insulinotropic peptide to their stimulus and action. LECTURE

238. **Phases of digestive control.** Describe the cephalic, gastric, and intestinal phases and state the dominant control mechanism in each. LECTURE

239. **Gastric secretion and mucosal protection.** Trace the production of hydrochloric acid by the parietal cell and explain how the mucosal barrier protects the stomach and how ulcers develop when it fails. LECTURE

240. **Pancreatic and biliary secretion.** State the components of pancreatic juice and bile and explain how bicarbonate neutralizes chyme and how bile salts emulsify fat. LECTURE

241. **Carbohydrate and protein digestion and absorption.** Trace a starch and a protein from mouth to bloodstream naming the enzymes at each step and the transport mechanism at the brush border. LECTURE LAB

242. **Lipid digestion and absorption.** Trace a triglyceride through emulsification, lipase action, micelle formation, and chylomicron packaging to the lymphatic route and explain why fat takes a different path than glucose. LECTURE

243. **Liver function and enterohepatic circulation.** State the metabolic, storage, and detoxification functions of the liver and trace the enterohepatic circulation of bile salts. LECTURE

244. **Large intestine and microbiome.** Describe water and electrolyte absorption in the colon, the contribution of the microbiome, and the defecation reflex. LECTURE

245. **Digestive enzyme experiment.** Test the effect of pH, temperature, and substrate on a digestive enzyme and relate the results to the region of the tract where that enzyme works. LAB

METABOLISM AND ENERGY BALANCE

246. **ATP production pathways.** Compare glycolysis, the citric acid cycle, and oxidative phosphorylation by location, oxygen requirement, and ATP yield. LECTURE

247. **Absorptive state.** Describe the fate of glucose, amino acids, and fats in the absorptive state and identify insulin as the dominant hormone. LECTURE

248. **Postabsorptive state.** Explain how glycogenolysis, gluconeogenesis, lipolysis, and ketogenesis maintain plasma glucose during fasting and name the hormones that drive each. LECTURE

249. **Integrated glucose regulation.** Predict the hormonal and metabolic response to a carbohydrate meal and to a prolonged fast and explain how plasma glucose stays within range in both. LECTURE LAB

250. **Energy balance and metabolic rate.** Define basal metabolic rate, state the factors that raise and lower it, and explain how indirect calorimetry estimates energy expenditure. LECTURE LAB

251. **Thermoregulation.** Trace the negative feedback response to heat and to cold through the hypothalamus and the effectors and explain the mechanism of fever. LECTURE

252. **Metabolic rate measurement.** Estimate metabolic rate from measured or simulated oxygen consumption and compare resting values with values after activity. LAB

IMMUNE PHYSIOLOGY

253. **Innate defenses.** Describe the physical, chemical, and cellular innate defenses and explain what makes them nonspecific and immediate. LECTURE

254. **Inflammation and fever.** Order the events of the inflammatory response and explain how the cardinal signs arise and why fever aids defense. LECTURE

255. **Adaptive immunity.** Compare humoral and cell mediated immunity by the cell responsible, the target, and the mechanism of elimination. LECTURE

256. **Antibody structure and function.** Relate antibody structure to antigen binding and name the mechanisms by which antibodies neutralize or mark a target. LECTURE LAB

257. **Immune memory and immunization.** Compare the primary and secondary antibody responses and explain how active and passive immunization produce protection. LECTURE

258. **Immune dysfunction.** Classify allergy, autoimmunity, and immunodeficiency by what the immune system is doing wrong in each. LECTURE

REPRODUCTIVE PHYSIOLOGY

259. **Hypothalamic pituitary gonadal axis.** Trace the gonadotropin releasing hormone axis to the gonads and place the feedback loops for both sexes. LECTURE

260. **Male reproductive physiology.** Explain the control of spermatogenesis by follicle stimulating hormone, luteinizing hormone, and testosterone and state the roles of the Sertoli and Leydig cells. LECTURE

261. **Ovarian cycle.** Order the follicular, ovulatory, and luteal phases and explain how the luteinizing hormone surge is triggered by positive feedback. LECTURE LAB

262. **Uterine cycle.** Correlate the menstrual, proliferative, and secretory phases of the uterine cycle with the estrogen and progesterone levels that drive them. LECTURE LAB

263. **Pregnancy and placental hormones.** Explain how human chorionic gonadotropin rescues the corpus luteum and state the roles of placental estrogen, progesterone, and relaxin. LECTURE

264. **Parturition and lactation.** Explain labor as a positive feedback loop driven by oxytocin and compare the hormonal control of milk production with milk ejection. LECTURE

265. **Hormone cycle graph interpretation.** Read a graph of gonadotropin and ovarian hormone levels across a cycle and identify the day of ovulation and the event driving each peak. LAB

INTEGRATION

266. **Multisystem integration case.** Given a clinical scenario, trace the disturbance through at least three organ systems and identify the compensatory responses in the order they occur. LECTURE LAB

267. **Exercise as an integrated response.** Build an integrated account of moderate exercise across the muscular, cardiovascular, respiratory, endocrine, and renal systems. LECTURE LAB

268. **Drawing based synthesis check.** Draw and annotate a homeostatic pathway from memory and explain each step aloud or in writing without notes. LECTURE LAB